

Which Quadrilateral Property Implies Which?

Answer Key

High School Geometry

Quick Answers

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|---|---|
| 1. (a) $x = 16$, — | — |
| 2. (a) $a = 4$, — | 5. (a) $t = 4$, (b) $PS = 14$, — |
| 3. (a) side length $AB = 15$, — | 6. (a) diagonal $AC = 9.85$, (b) diagonal $BD =$ |
| 4. (a) $y = 5$, (b) length of each diagonal $= 26$, | 5, — |
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What is verified

9 of 15 answers machine-checked against SymPy, across 6 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-CO.C – problems 1, 2, 3, 4, 5, 6

Difficulty 2–5 of 5 across 15 tagged checks.

Common wrong answers

- If they got 15: assumed a parallelogram must have right angles and set the first angle equal to 90*
 - If they got 21: added the two half-diagonals instead of using the pythagorean theorem*
 - If they got 2.4: read bisect each other as each diagonal being half of the other*
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1. Parallelogram $ABCD$: $m\angle A = (5x + 15)^\circ$, $m\angle B = (3x + 37)^\circ$. Kim sets $5x + 15 = 90$ and gets $x = 15$.

Solution (a): $\angle A$ and $\angle B$ are *consecutive* angles of a parallelogram, so they are supplementary — that is the property a parallelogram actually guarantees.

$$(5x + 15) + (3x + 37) = 180$$

$$8x + 52 = 180$$

$$8x = 128$$

$$x = 16$$

So

$x = 16$

Check: $m\angle A = 95^\circ$, $m\angle B = 85^\circ$, and $95 + 85 = 180$. Neither is a right angle, which is exactly why Kim's assumption fails.

Solution (b) — instructor-judged. A model response: every rectangle is a parallelogram, but not every parallelogram is a rectangle. Kim used the implication in the wrong direction: knowing $ABCD$ is a parallelogram tells you the opposite angles are congruent and the consecutive angles are supplementary — it does not give you a right angle. She should have used the supplementary-consecutive-angles property.

Full credit: states the implication only runs rectangle $\Rightarrow$ parallelogram, and names the supplementary-angle property as the correct tool. *Half credit:* names the correct property but not the reversed implication. *No credit:* redoes the algebra with no account of the faulty assumption.

2. Quadrilateral $JKLM$: $\overline{JK} \parallel \overline{LM}$, $JK = LM$, with $JK = 3a + 5$ and $LM = 7a - 11$.

Solution (a):

$$3a + 5 = 7a - 11$$

$$16 = 4a$$

$$a = 4$$

So

$a = 4$

and each of those sides measures 17.

Solution (b) — instructor-judged. A model response: one pair of opposite sides that is both parallel and congruent is enough to force a parallelogram. It is not enough to force a rectangle: a rectangle needs one further condition — a right angle, or congruent diagonals — and nothing in the problem supplies one. The figure could be strongly slanted and still satisfy everything given.

Full credit: names the missing extra condition (right angle or congruent diagonals). *Half credit:* says it could be slanted without naming the condition. *No credit:* claims it is a rectangle.

3. Rhombus $ABCD$ with diagonals meeting at E : $AE = 9$, $BE = 12$. Find AB , then judge the square claim.

Solution (a): The diagonals of a rhombus are perpendicular bisectors of each other, so $\triangle AEB$ has a right angle at E with legs 9 and 12.

$$AB = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$$

So

$AB = 15$

Solution (b) — instructor-judged. A model response: all four sides being congruent is the *definition* of a rhombus, which is what was given in the first place — it proves nothing new. Every square is a rhombus, but not every rhombus is a square. To call $ABCD$ a square you would also need right angles at the vertices, equivalently congruent diagonals; here the diagonals are 18 and 24, which are not congruent, so it is definitely not a square.

Full credit: states the square $\Rightarrow$ rhombus direction and names the missing right-angle (or congruent-diagonal) condition. *Half credit:* rejects the claim with no missing condition named. *No credit:* agrees with the classmate.

4. $WXYZ$ has congruent diagonals bisecting each other, $WY = 6y - 4$, $XZ = 2y + 16$. Ravi reports $y = 2.4$.

Solution (a): “The diagonals bisect each other” says each diagonal is cut in half by the other, not that one diagonal is half the other. The diagonals are also given as congruent, so set the two expressions equal:

$$6y - 4 = 2y + 16$$

$$4y = 20$$

$$y = 5$$

So

$$y = 5$$

Solution (b): $WY = 6(5) - 4 = 30 - 4$, and $XZ = 2(5) + 16 = 10 + 16$ agrees.

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Solution (c) — instructor-judged. A model response: diagonals that bisect each other make $WXYZ$ a parallelogram. Congruent diagonals then upgrade a parallelogram to a rectangle, so $WXYZ$ is a rectangle. A rectangle is a square only when the sides are congruent as well, and nothing given establishes that — Ravi ran the square $\Rightarrow$ rectangle implication backwards.

Full credit: reaches rectangle by the two diagonal properties and explains why square does not follow. *Half credit:* reaches rectangle only. *No credit:* concludes square.

5. Kite $PQRS$: $PQ = QR$, $PS = SR$, $PQ = 4t - 1$, $QR = 2t + 7$, $PS = 3t + 2$.

Solution (a): The kite’s first pair of adjacent sides is congruent:

$$4t - 1 = 2t + 7$$

$$2t = 8$$

$$t = 4$$

So

$$t = 4$$

Solution (b): $PS = 3(4) + 2 = 12 + 2$.

$$PS = 14$$

Solution (c) — instructor-judged. A model response: the first pair measures $4(4) - 1 = 15$ and the second pair measures 14. A rhombus requires all four sides congruent; here two sides are 15 and two are 14, so this figure fails that test. Every rhombus is a kite — a rhombus just happens to have both pairs of adjacent sides congruent — but the implication does not run the other way.

Full credit: compares 15 with 14 and names the all-four-sides-congruent condition. *Half credit:* notes the lengths differ without naming the condition. *No credit:* calls it a rhombus.

6. $ABCD$ with $A(0, 0)$, $B(6, 0)$, $C(9, 4)$, $D(3, 4)$; $AB = DC = 6$ and $AD = BC = 5$.

Solution (a): $AC = \sqrt{(9 - 0)^2 + (4 - 0)^2} = \sqrt{81 + 16} = \sqrt{97} = 9.8488\dots$, so

$$AC \approx 9.85$$

Solution (b): $BD = \sqrt{(3 - 6)^2 + (4 - 0)^2} = \sqrt{9 + 16} = \sqrt{25}$, so

$$BD = 5$$

Solution (c) — instructor-judged. A model response: both pairs of opposite sides are congruent, which forces $ABCD$ to be a parallelogram. A rectangle additionally requires

congruent diagonals, and $AC \approx 9.85$ while $BD = 5$ — nowhere near equal — so $ABCD$ is not a rectangle. Being a parallelogram never implies being a rectangle; the implication runs only the other way.

Full credit: uses both diagonal lengths and names the congruent-diagonal test for a rectangle. *Half credit:* compares diagonals with no test named. *No credit:* answers from the picture alone.